Shenghan **Gao**

Photonic Integrated Circuits · Photonics Hardware · Validation & Test

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Summary

Ph.D. candidate in Electrical and Computer Engineering specializing in integrated photonics, photonic hardware validation, and coherent electro-optic systems. End-to-end experience across silicon photonics and TFLN/LNOI PIC design, passive and active photonic devices, cleanroom fabrication, fiber-to-chip integration, RF/EO characterization, DSP-based signal analysis, and Python/PyVISA/SCPI automated measurement. Seeking PIC design, silicon photonics, photonics validation/test, electro-optic device, and optical hardware R&D roles.

Core Skills

PIC design	Silicon photonics and TFLN/LNOI PIC design; waveguides, directional couplers, MMI/splitters, grating/edge couplers, interferometers, electro-optic modulators, thermal phase shifters
Simulation/layout	GDSFactory, Cadence Virtuoso, KLayout; Lumerical, RSoft, COMSOL, Zemax/CodeV; optical mode analysis, device optimization, PIC layout preparation
Fab/integration	Cleanroom fabrication, spin coating, photoresist processing, electron-beam lithography, etching, SEM inspection; fiber-array coupling, fiber-to-chip coupling, photonic packaging, optical alignment
Validation	Device characterization, high-speed EO/RF characterization, VNA, RF detection, high-speed/balanced photodetection, EOMs, OSAs, SVAs, spectrum analyzers, oscilloscopes, tunable lasers
Automation/DSP	Python measurement automation, PyVISA/SCPI, NI MAX, automated testing and data acquisition, parameter sweeps, data logging, FFT/frequency-domain analysis, DSP equalization, MATLAB
Coherent optics	Optical injection locking, semiconductor-laser rate equations, coherent interference, carrier recovery, AM/PM modulation transfer, PID phase stabilization

Research Experience

University of Pittsburgh — *Pittsburgh, PA*

Research Assistant | Advisor: Nathan Youngblood — *Spring 2020 - Present*

- **Distributed coherent photonic computing:** Developed an optical-injection-locking framework for coherent optical processing across distributed photonic nodes, combining semiconductor-laser rate-equation modeling with experimental validation.
- Mapped injection-ratio and frequency-detuning operating regions; analyzed carrier recovery and modulation transfer using coherent detection, parameter sweeps, and FFT/DSP-based signal analysis.
- Built coherent EO/RF testbeds with EOMs, high-speed and balanced photodetection, optical and RF spectrum analysis, lock-in detection, PID phase feedback, and fiber-stretcher control.
- **PIC design and integration:** Designed chip-scale photonic circuits to translate fiber injection-locking systems onto integrated platforms, including waveguides, directional couplers, MMI/splitters, grating couplers, edge-coupling interfaces, and interferometric paths.
- Modeled silicon photonics and TFLN/LNOI passive and active devices using Lumerical, RSoft, and COMSOL; developed layout workflows with GDSFactory, Cadence Virtuoso, and KLayout for optical mode analysis, electro-optic modulation, and device optimization.
- Performed and supported cleanroom fabrication and process evaluation, including spin coating, photoresist processing, e-beam lithography, etching, and SEM inspection; aligned fabricated chips using fiber arrays, grating coupling, and edge coupling.
- **Automated hardware validation:** Developed Python/PyVISA/SCPI and NI MAX workflows for wavelength, power, injection-ratio, and frequency sweeps; connected device measurements to FFT/DSP analysis, plots, and repeatable reports.

University of Rochester — *Rochester, NY*

Research Assistant | Advisor: Xi-Cheng Zhang — *Sep 2017 - Aug 2019*

- **Liquid-plasma THz generation:** Investigated terahertz emission under ultrashort optical excitation; characterized how liquid temperature, polarity, and viscosity affected emission strength, bandwidth, and spectral content.
- Analyzed time- and frequency-domain measurements using SNR, sensitivity, regression, and repeatability analysis to relate liquid properties to THz performance.
- **Digital cloaking:** Prototyped a lenslet-array imaging system with MATLAB pixel remapping to reconstruct optical images.

Education

University of Pittsburgh

Ph.D. in Electrical and Computer Engineering — *Dec. 2026 (expected)*

Georgia Institute of Technology

M.S. in Computer Science, Machine Learning track — *May 2027 (expected)*

University of Rochester

M.S. in Optics — *Aug. 2019*

University of Rochester

B.S. in Optical Engineering | B.S. in Applied Mathematics — *Aug. 2017*

Industry Experience

Lumentum Operations LLC *San Jose, CA*

Research Intern | Mentor: Joe Wen *May 2023 - Aug 2023*

- Developed MATLAB/Python FIR-filter equalization workflows for communication signals distorted by intersymbol interference, connecting DSP filter design to recovered waveform quality.
- Collected measured signal data and applied FFT/frequency-domain analysis to compare filter responses, tune coefficients, and validate demodulated-signal recovery across test cases.

Sunny Optics *Ningbo, China*

Research Intern | Mentor: Chao Xu *May 2018 - Aug 2018*

- Designed and optimized automotive LiDAR and mobile-camera lens systems in CodeV/Zemax against optical performance, tolerance, and packaging requirements.
- Collaborated with opto-mechanical engineers during lab integration, using alignment and test feedback to refine optical system functionality.

Selected Publications and Talks

Distributed Coherent Optical Computing via Injection-Locked Photonic Networks

Gao, S., K. Lüdge, F. Da Ros, N. Youngblood, submitted to APL, under review, 2026

Distributed Coherent Photonic Computing via Injection Locking

Gao, S., SPIE Optics + Photonics (oral presentation), 2026

Distributed Coherent Optical Computing via Injection-Locked Photonic Networks

Gao, S., Invited PhD Candidate Talk, Pittsburgh Quantum Institute, Pittsburgh, 2025

Strain-free FBG sensors for accurate temperature measurements from cryogenic temperatures to 300 K

Gao, S., P. R. Swinehart, S. Zhong, J. Wuenschell, N. Lalam, M. Buric, K. P. Chen, SPIE Defense + Commercial Sensing (oral presentation), 2023

Preference of subpicosecond laser pulses for terahertz wave generation from liquids

Q. Jin, Y. E., **S. Gao**, X.-C. Zhang, *Advanced Photonics* 2(1), 015001, 2020

Investigation of liquid properties on emitting terahertz wave under ultrashort optical excitation

S. Gao, Q. Jin, Y. E., X.-C. Zhang, SPIE, 2019